

BSc Hons in Computing

2024.2

**An Explainable Multi-Source Spatiotemporal Model for
Traffic Accident Hotspot Prediction and Severity
Classification in Sri Lanka**

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Chapter 1: Introduction

1.1 Background

Road traffic accident cause significant damage to human lives and economy each year around the world. According to the World Health Organization approximately 1.19 million loses their live annually while also causing significant injuries and economic damages (World Health Organization, 2023). As a country Sri Lanka also records a gradual increase in accidents each year due to factors like urbanization and increased number of vehicle usage. According to the Police Spokesman ASP F.U. Wootler, the number of deaths caused by accidents has increased by 322 compared to 2024, recording 2710 death in the year 2025 (Administrator, 2026).

In order to prevent the negative impact of traffic accidents, Sri Lankan government has taken necessary measure which include introduction of new rules and regulations, policies, awareness programmers and many more (Ministry of Transport, n.d.). But still there is an increase in number of accidents in recent years due to factors like road poor road conditions, reckless driving and excessive speeding (Over 670 Fatal Road Accidents Reported so Far This Year - Sri Lanka Police, 2026). This shows that Sri Lanka needs data driven predictive strategies to reduce the impact of traffic accidents

Accordingly, this research proposes developing of a multi-source machine learning framework for accident risk prediction and spatial analysis in Sri Lanka, which will be integrated with Sri Lankan police accident records and contextual weather data. This framework will help decision making process of the relevant authorities by identifying accident hotspot locations, contributing factors for the accident prediction, dynamic variations across accidents hotspots over a period of time and presenting those information via an interactive dashboard.

1.2 Research Problem

As it is mentioned earlier, traffic accidents cause harm to human lives and the economy both around the world and Sri Lanka. Even though Sri Lankan government has taken necessary steps like traffic laws, fines, and signal systems to reduce the impact, the statistics shows that the number of traffic accidents fluctuates each year with a noticeable increase in traffic accidents in recent years (Department of Census and Statistics, 2024).

Researchers have explored using machine learning techniques for accident hotspot prediction and analysis. But most of these studies are geographically limited to their own country. For example, by using supervised machine learning techniques, researchers have developed a framework for accident hotspot prediction on Brazilian highways using multiple data sources such as location, time and weather conditions (de et al., 2023). But such kind of frameworks are not validated on Sri Lankan context where factors like weather and traffic conditions are significantly differ from those countries.

When it comes to Sri Lankan context only a limited number of studies have been done on this exact domain. For example, Jayamanna and Kalansooriya (2024) has explored on developing a machine learning based hotspot prediction system. However, only a limited number of features from a single source (historical police accident) has been used for this research. But some studies show that integrating contextual data such as weather data can improve model's prediction performance (Yan & Shen, 2022). Accordingly, there is a need for developing a machine learning framework with multiple data sources.

Accordingly, Sri Lanka needs a machine learning framework that uses multiple data sources such as police accident data and weather data for accident hotspot prediction and analysis of spatial temporal features.

1.3 Objectives of the Project

- To collect, preprocess and integrate police accident records with contextual weather data into a centralized data warehouse via an ETL pipeline.
- To capture dynamic accident risk variations across regions and time by constructing spatiotemporal features.
- To develop and compare performance of multiple machine learning models such as Logistic Regression, Random Forest and XGBoost to identify the most suitable model when using multiple data sources to predict accident hotspots and classify severity in Sri Lanka.
- To analyze the contribution of spatial and temporal features such as location, time and weather conditions to the accident hotspot prediction and severity prediction using XAI techniques such as SHAP
- To develop a prototype dashboard with necessary API for visualization of predicted hotspot location along with location based filtering and meta data insights that support decision making.

1.4 Research Questions

- How to collect, preprocess and integrate police accident records with contextual weather data and store in a data warehouse via an ETL pipeline?
- How to capture accident risk variation across regions and time using spatiotemporal features such as time patterns and location based grids?
- What is the best machine learning model among Random Forest, Logistic Regression, XGBoost, when it comes to accident hotspot prediction and severity classification in the Sri Lanka using multiple data sources?
- To what extent do the spatial and temporal features such as location, time and weather conditions, contribute to the prediction performance of the model?
- How can a dashboard and an API be developed to visualize the predicted hotspot location along with location-based filtering and metadata insights to support decision making?

1.5 Scope of the research

The main objective of the research is to develop a multi source machine learning framework for accident hotspot prediction and spatial risk analysis in Sri Lanka. For the implementation, historical police accident records along with weather data obtained through external APIs based on accident location, time and data will be used. Also the scope of this research will be limited to a selected set of regions due to data scarcity in Sri Lankan context.

The inputs for this research include accident related data such as location, date and time , severity, vehicle involvement, casualties, and cause of the accident along with the contextual weather data, which includes features like precipitation, visibility and temperature. Once collected, these data will go through an ETL pipeline where preprocessing techniques like extraction, transformation, feature engineering and finally integrating to a centralized data warehouse happen. After this, these data will be used to train and evaluate machine learning models such as random forest, Logistic Regression and XGBoost to identify the most suitable for Sri Lankan context.

The final output of the research will be a prototype dashboard and an API with only core functionalities needed for visualization of predicted hotspot location along with location based filtering and meta data insights that support decision making.

Additionally, this framework is developing for authorities such as traffic police authorities and road safety analyst who can use the system to identify hotspot prone areas, contributing factors for predicted accidents and thereby to support data driven decision making.

Also, it is very important to note that this research will be limited only to historical data of selected regions as there is no any systematic approach to obtain real time or livestream accident data within Sri Lanka.

Chapter 2: Literature Review

2.1 Findings by other researchers

Road traffic accident prediction has become an important research area due to the increasing number of traffic accidents with the rapid urbanization in recent years. Researchers have already explored applying various machine learning techniques for hotspot prediction and analysis

2.1.1 Machine learning approach to traffic accident analysis and hotspot prediction

This study focuses on traffic accident analysis and accident hotspot prediction in the Setúbal district of Portugal. A combination of both supervised and unsupervised models such as Decision Trees, Random Forest, Logistic Regression, Naïve Bayes, Hierarchical Clustering, and DBSCAN has been used for hotspot prediction. Also, according to the findings of the research, random forest performed best when it comes to hotspot predation and DBSCAN was effective when it comes to identify accident prone clusters (Santos et al., 2021).

2.1.2 Machine learning approach for classifying road accident hotspots

For this, several supervised machine learning models such as Support Vector Machine (SVM), Random Forest, XGBoost, and Artificial Neural Networks (ANNs) have been used. Also this shows that incorporating external variables such as weather data can improve the model's performance significantly. However, such approaches remain unexplored when it comes to Sri lanka where weather patterns and peak hours significantly differ from those countries (de et al., 2023).

2.1.3 Sri Lankan Accident Hotspot and Severity Prediction Model

This study has explored on developing a machine learning based mobile application for accident hotspot prediction and severity classification. Accordingly, XGBoost for hotspot regression and Random Forest for severity classification have been used. However, this study was conducted using limited number of features that was obtained through a single source (historical police accident records) and was limited to the Mount Lavinia region (J.M.C.T. Jayamanna & P. Kalansooriya, 2024).

2.1.4 Explainable AI traffic accident analysis and hotspot prediction

These researchers developed an interpretable machine learning model using Random Forest and SHAP to identify factors that contributing most to the accident severity predictions. Accordingly, it is found that speed contributed most for the severity prediction. These study shows importance of XAI when it comes to improving transparency of the model and thereby helping to make evidence based decisions. However, such frameworks are unexplored when it comes to Sri Lanka (Al-Omari et al., 2025).

2.1.5 Spatiotemporal Risk Mapping of Statewide Weather-related Traffic Crashes: A Machine Learning Approach

For this research models like XGBoost with spatiotemporal risk mapping techniques such as space-time cube analysis, DTW, and Getis-Ord Gi*, has been used to identify weather related traffic accident hotspots in Nort Carolina region of USA. The study shows that spatial and temporal dependencies can improve model's prediction accuracy, showing importance of identifying dynamic accident risk variations between space and time. However, this research is only limited to weather related crash data, making it less applicable to generalized traffic conditions and regions (Ogungbire & Pulugurtha, 2025).

2.2 The research gap

According to some of the researches, it is clear that integrating traffic accident data with contextual data such as weather data can significantly improve the prediction performance of the model (Ogunbire & Pulugurtha, 2025; Santos et al., 2021). But when it comes to Sri Lanka, most of the studies rely only on single data source such as historical police accident record. Existing studies related to hotspot analysis in Sri Lanka also suggest that integrating weather related data can have significant impact to hotspot identification and analysis process (Nalaka et al., 2021). Accordingly, Sri Lanka needs a centralized data warehouse integrated with multiple data sources such as police accident records and weather data.

Also when it comes to real world accident occurrence varies across different time periods due to factors like traffic hours, and weather and such factors contributed greatly to the model's performance prediction (Yan & Shen, 2022). Also according to the study by Ogunbire & Pulugurtha (2025), it is found that spatial and temporal dependencies can improve model's prediction accuracy. However, this research is only limited to weather related crash data and only a limited number of research has been done on capturing those spatiotemporal variations using features such as time patterns and location based grids, when it comes to broader accident hotspot prediction. Accordingly, a gap remains in combining spatial and temporal features of accidents to create a model that can perform more context aware predictions.

Additionally, most of the existing systems in Sri Lanka were given lesser focus to feature contribution and model explainability analysis. Even though predicting a traffic accident itself is very important, authorities need to know why the model considers a certain location as high-risk hotspot, to support decision making. But machine learning models act as black box systems, limiting their adaptability in safety critical areas. However, it is possible to apply Explainable AI techniques such as SHAP to identify contribution of each feature to the prediction score and thereby it is possible to improve the model's interpretability (Hamdan & Sipos, 2025; Omar Ibrahim Aboulola, 2024; Islam et al., 2022).

But most of the existing research in Sri Lanka has not explored the contribution of features such as location, weather, accident timing and more to predicted accident using explainable AI techniques such as SHAP. Therefore, a gap remains when it comes to model's interpretability, which is very important for decision making.

Based on above gaps, this study proposes a multi-source explainable spatiotemporal machine learning framework for accident hotspot prediction and severity classification for Sri Lankan context. Also, unlike exiting studies, it is planned to integrate both police accident records and weather data, thus creating a multi source data warehouse. Additionally spatiotemporal analysis via feature engineering to identify variations across accident hotspots and explainable AI techniques such as SHAP will be integrated to improve the model's prediction performance and interpretability.

2.3 Summary

Author	Key Finding	Research Gaps
Santos et al. (2021)	• Random Forest performs best. • Use multi-Source Data, • Clustering for hotspot prediction	• No XAI techniques like SHAP • Limited temporal analysis
de et al. (2023)	• ANN Forest performs best among SVM, RF and ANN. • Limited spatiotemporal analysis	• Limited explainability • Limited spatiotemporal analysis
J.M.C.T. Jayamanna & P.	• XGBoost performs best for hotspot prediction	• Single data source • No XAI techniques • Limited features

Kalansooriya (2024)	• Random Forest performs best for severity classification	• No spatiotemporal modelling
Al-Omari et al. (2025)	• Random Forest performs best. • Spatial and temporal pattern analysis • Feature Importance analysis via SHAP	• Limited to Irbid region of Jordan and not validated against Sri Lankan context
Ogungbire & Pulugurtha (2025)	• Spatiotemporal analysis for risk detection.	• Only limited to weather related crash data.

2.4 Chapter Conclusion

In conclusion, several machine learning approaches have been taken to predict accident hotspots and it is also highlighted that integrating factors such as weather can improve the prediction performance of models.

However, when it comes to the Sri Lankan context, no research has incorporated multiple sources such as weather data and the usage of XAI techniques such as SHAP and spatiotemporal analysis. Accordingly, there is a need for a multi-source explainable spatiotemporal machine learning framework to predict accidental hotspots and classify severity of them.

Chapter 3: Methodology

This research will follow a data driven methodology to develop a multi source, explainable spatiotemporal machine learning framework for accident hotspot predication and severity classification in Sri Lanka.

3.1 Data Collection

The primary data source for this research will be anonymized police accident records which is planned to be obtain from Sri Lankan police and it is planned to extract primary feature such as location, date and time, casualties, vehicle types, number of vehicles, cause of accidents, accident severity and etc. However the data collection will be limited to few accident prone areas due to data scarcity in Sri Lanka. As the secondary data source, weather data will be collected from an external weather API for given accident location, data and time as it can improve the accident hotspot prediction performance of the model(Ogungbire & Pulugurtha, 2025; Santos et al., 2021).

3.2 Data Preprocessing and Spatiotemporal Feature Extraction

The collected data will then be processed via an ETL pipeline where Extraction, transformation and loading of data will be happened. Also preprocessing techniques such as handling missing values, data cleaning and normalization will be applied on data during transformation stage. Once this done, feature engineering will be performed on data to extract spatiotemporal variables such as location based grids, day of week, peak hour indicators and rainfall categories as these features can be used to improve model's prediction performance and also help to understand dynamic risk variation across accident hotspots (Ogungbire & Pulugurtha, 2025). Once this is done, preprocessed data then will be stored on a centralized Datawarehouse in order to facilitate model training.

3.3 Machine Learning Modelling and Evaluation

For this study, supervised machine learning approaches such as random forest, Logistic Regression and XGBoost will be incorporated for accident hotspots prediction and severity classification due to their effectiveness in handling structured tabular data and maintaining

interpretability through SHAP analysis (Hamdan & Sipos, 2025). Once the models are trained and developed, model evaluation metrics such as accuracy, precision, recall and F1 Score will be used to evaluate the above-mentioned machine models to find the most suitable model for Sri Lankan context.

3.4 Explainable AI Approach

Once the most suitable model for accident hotspot prediction and severity classification is identified, explainable AI techniques such as SHAP can be integrated to interpret the model's prediction (Islam et al., 2022). By doing this, it is possible to identify contribution of spatiotemporal feature that contributed to accident hotspot prediction, supporting decision making process.

3.5 Development of dashboard and API

As the final step a prototype dashboard along with an API will be developed to visualize the predicted accident hotspots and severity classification along with necessary feature contribution extracted via machine learning model.

3.6 Architecture diagram for the Proposed Framework

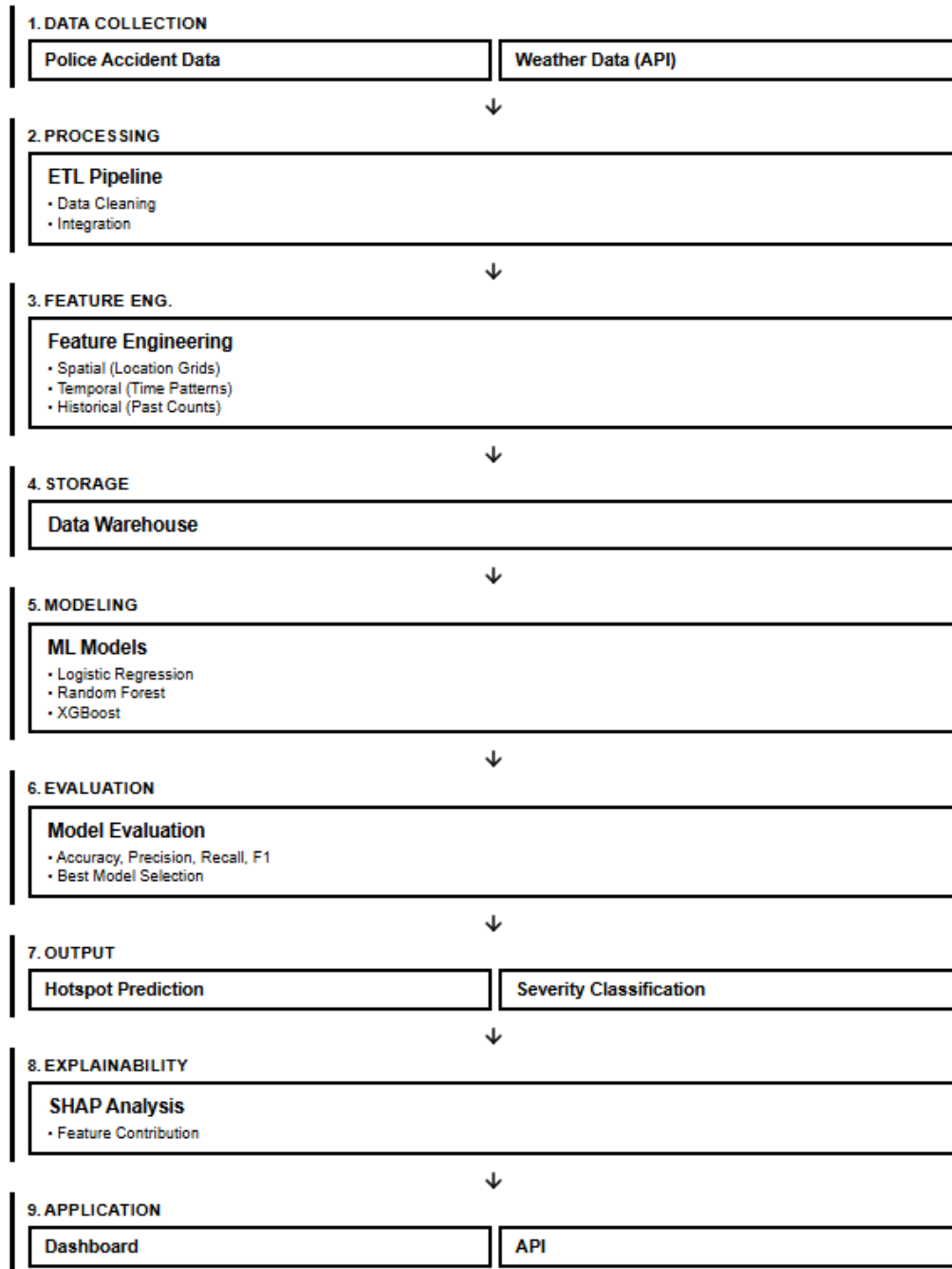


Figure 1 Architecture Diagram

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Appendix

Workplan / Timeline

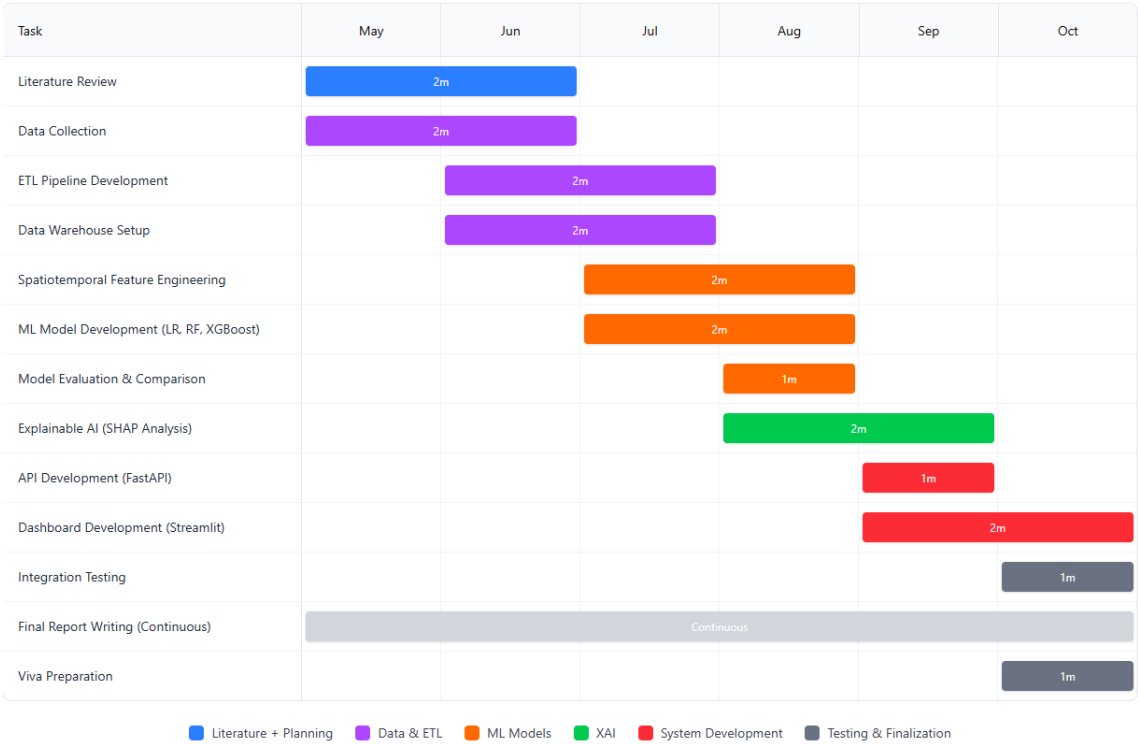


Figure 2 Gantt Chart -Work Plan

Progress Monitoring Sheet

BSc Hons Computing/Information Technology for Business/Ethical Hacking and Network Security

School of Computing and Engineering

National Institute of Business Management

Progress Monitoring Sheet

Final Project(Proposal)

Degree Title : BSc (Hons) Computing (Software Engineering Pathway)
 Student CU Index : 16114760
 Research Title : An Explainable Multi-Source Spatiotemporal Model for Traffic Accident Hotspot Prediction and Severity Classification in Sri Lanka
 Supervisor Name : Mr. Niranga Dharmaratna

Meeting No	Meeting Date	Supervisor Comments	Supervisor Signature
01	23/03 2026	Discussed some pre-liminary thoughts on the project core. Some suggestions were given to tweak it further to fit in to the module requirements.	
02	31/03 2026	Further discussion att on the topic depth and scope. —	

03	09/04 2026	Further issues on conceptual model. Work flow enhancements.	
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04	16/04 2026	Peer review and feedback.	
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Supervisor Final Comments and Signature:



Date : Apr 19, 2026

Figure 3 Progress Monitoring Sheet